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# Global Market Research

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Prof. Wendy Olsen

## Permuting and Combining Elements in Words or Itemsets

The reason we learn to compute **how many unique combinations exist** for an itemset is to gain an understanding of the order of magnitude of a market-research problem. The specific application we will use is information flows. The huge size of information flows – at least as a potential size – does have a limit, and the combinatorial algebra helps to estimate that limit. An information flow in market research might be a monthly delivery of a full database of all the transactions of a firm. Because it's monthly, we need to quickly decide how to organise and analyse the data.

First note that an information flow is a series of events, or signals. We use symbols to represent events. We use the binary number system (eg 10011011) to represent the integers going from 1 upward, and this information flow is shown right-to-left as (0 or 1) repeating. A repetition of such a flow has $2^n$ combinations for n digits. The 2*2*2*...2 reasoning matches that shown in the 'combinations' algebra, which appears below. But first, we consider more challenging situations where a single alphabetic word is permuted k times. For example, if we permute abc in all its possible sequences, we get: abc, acb, bac, bca, cab, cba. There are six permutations. Learn the permutations formula and apply it to that question.

P(3,1)*P(2,1)*P(1,1) Eq. 1

is one way to express this. And so, 3*2=6. Or, write

P(3,3) Eq. 2

and this becomes 3!/0! = 3*2 = 6. More on this below.

You will need the product rule, shown with * in equation 1. See details below.

Then learn the combinations formula, which can be used to prove why $2^n$ is the solution for binary number system. For instance, if we choose 1 from 2 symbols, n times for n digits: the result is 2*2*2... until you have taken one the product n times. That applies the product rule, here giving $2^n$. Altogether, the mathematics I have listed here is known as the combinatorial algebra.

# THE PERMUTATIONS ALGEBRA

Permutations are lists of characters or objects, which have a specific sequence and cannot have duplicates.

The general formula for permutations is P(m,n) = (m!)/((m-n)!)

$P(m,n) = \frac{m!}{(m-n)!}$ Eq. 3

First Anagram Task
The **permutations** of spelling of a name like Angelford include Angleford, Engalford, and Afgelnord. (A permutation with 8 letters is not allowed, by assumption, but if you need to calculate all permutations of *all lengths* you will **add** together the permutations available for each length.)

Be thoughtful: When we draw the first letter, we are drawing it from a basket that contains {a, d, e, f, g, l, n, o, r} . The list is an **itemset** because all the elements are unique.* We could have drawn from the itemset list with replacement, taking up to 9 letters, but that won't give us the particular anagrams we want here. We need to know the scale of the problem. Allowing duplicates and re-samples will tend to increase the scale of the problem.

Drawing with replacement, without attention to order, would involve the **combinations algebra** and other product rule operations.

*Notice an odd fact. Suppose our anagram began with Angelman rather than Angelford. Now the eight letters are not distinct. We have the actual set {a, a, e, g, l, m, n, n}. The corresponding itemset is {a, e, g, l, m, n} with 6 elements. An itemset by definition has unique elements.

We can describe the eight events in Angelman by using notation. Each event is a choice of one letter from the itemset {a,e,g,l,m,n} (choose 1 from six). You are welcome to pursue this example further.

Let us return to Angleford. I want to be clear about my anagram task. How many permutations are there of Angleford, which use the exact same letters but in different sequences?

Here we could say that each letter chosen for Angelford is a distinct event! (a fake assumption, but useful) - so we label them {1,2,3,4,5,6,7,8,9}. Now we are sure they are distinct, we can calculate how many permutations of these 9 events there are, in

the sequence. We will multiply together each count. The total length of the word is 9, so there are 9 parts.

(Note: answers are at the back of the handout)

Formula:

Insert your numbers:

Calculation:

Using R, check it:

Second Anagram Task:
Suppose now we can choose any distinct letter to go in each of nine spaces. How many permutations could exist, for this word game?
P(26,9) = 26!/(17!) = 26*25*24*23*22*21*20*19*18.

Eg. Bacfnpout.

The denominator is very simple in permutations. We now have 26*25*24*23*...*18. It is 1.133837e+12, an enormous number. It is 1,133,836,704,000.

Factorials are the product of k*(k-1)*(k-2)*... *(1). The factorial of 1 is 1.

Permutations can be specified to either allow or not allow for re-drawing a letter, by using my trick about labelling the events versus the actual letters. Thus in the game 'Scrabble', you can draw 7 letters, but you cannot replace once you start working from the 7 letters. The 'hand' you draw can include duplicates because the letter-bag contains lots of duplicate letters!

Hypothetical Q1: How many different scrabble hands can be received, each is 7 letters without replacement from a large group of letter-types? It depends on the group of letter-types. This question is relevant to loading containers onto a train, with several duplicate container-contents in the storage area. It's complex.
Hypothetical Q2: How many different words could you form, ideally, from your 7 letter hand? You get 7 and place either 2, 3, ... or 7 letters onto the board as a word.
So for 2-letter words, there are P(7,2); For 3-letter words, there are P(7,3);
For 4-letter P(7,4)... 5-letter words P(7,5), .... 6-letter words P(7,6) and
For 7-letter words, P(7,7) which is 1. We add up these permutations, not multiply them, giving 42 + 42*5 + 42*20 + 42*60 + 7 + 1 = 42*86 + 8 = 3612 + 8 = 3620.
It's amazing that the scale of the mental problem for people playing Scrabble is so large. The analogy in market research is that if the space in an advertising medium holds 20 adverts per hour, and we have several adverts to place but we cannot duplicate them at a point in time, and then we rotate the adverts, how many advertisers can be charged and for how much time each? Eg 20 per hour? Or is it

much higher, since each advert will be displayed for about 30 seconds, giving... It is complex.

PERMUTATIONS rely on the sequence. Unlike the hand drawn in Scrabble, they do not allow duplicated symbols. But whilst the letters or symbols are distinct, an information event could repeat some signal.

For example in Scrabble, a hand could consist of G, H, E, E, I , Z, Q. Among all the 3620 sequences, you might get the single, unique best word. How clever we are, playing Scrabble! What is the best word? _____

COMBINATIONS do not rely on the letters' order or sequence, and do not allow duplicates. A combination is more like a list of product categories. We need to decide which categories to include, and which to exclude. But the categories are distinct and unique.

Be careful: You may need to combine small combinations using the product rule.
& Be careful: if you could allow combinations that are smaller than k categories, then you may need to **add up** all the possible combinations of each different length.

Combinations can be itemsets. But a list of unique combinations is more focused, and fewer, than the corresponding set of permutations.

## ITEMSET DEFINITION

An itemset is a list of distinct items. In the list, each one is unique.

## THE PRODUCT RULE

When we combine chunks of symbols, each chunk can be a 'cell' or group of cells, and we can make new combinations by rotating all possible chunks B with each chunk in list A. As a result, count(A)*count(B) is a simple product rule for combinations. The two choice acts have to be mathematically independent.

Non-independence would mean that if you choose A first, and that choice reduces the options in B, then a subtraction or change is needed in the B list to make it accurate.

If we are picking from abstract itemsets like {a,b,c,...z} and {1,2,3,...10} there is independence.

# THE COMBINATORIAL ALGEBRA

The License Plate Problem
You could have xxx alpha, nn numeric, x alpha, n numberic for a total of 7 characters. How many plates can be produced [which are unique]? [these choices reflect independent groups since the use of the symbols is unlimited]

C(26,3)*C(10,2)*C(23,1)*C(9,1)

Obviously the last two characters allow *23 and *9 unique new plate options. Each * represents the use of the product rule.

Use R to solve this.

In reality, the 7-symbol plates are each distinct, but within them, some repeating of letters or numbers can occur. However, we aren't allowing that here. **'Sampling Without Replacement'** is the word for not allowing repeats of letters. That's what we have asserted.

The general formula for combinations is C(n, k) = $\binom{n}{k} = \frac{n!}{(k!)(n-k)!}$ Eq. 4

We say "26 choose 3" and write $\binom{26}{3}$. It follows the `26 choose 3' rule, formula: 26!/(23!3!) or 26*25*24/(3*2), thus, the number of unique 3-digit prefixes is 2600. Amazingly high.

Overall the number of unique license plates is 24,219,000.

The excel spreadsheet might convert this to scientific notation (format general), but you can convert it back using Format Number. All combinations are integers.

In R, you can insert numbers into equations using scientific notation using ^ to refer to exponentiation.

If we sold 89,000 plates, how many do we have left?
Result<- 24,219,000-89000
Print(paste("We now have ",result, "plates left."))
We now have 24,130,000.0 plates left.

Query: Is the list of 24,219 thousand license plates an itemset? _____
Use caution about the " symbol. In R, the expression in () above will require straight double-quote marks, found on the keyboard as SHIFT-2 when R is editing. But when Word is editing, it presents curved double quote marks which do not translate well into R scripts.

The use of print and paste is to ensure that a combination of strings and algebra is seamlessly presented. The paste function converts algebra to strings. The print function displays the contents of the ().

Query: Think back to the anagram task. Which is larger, the permutations or the combinations of the letters in Angleford? !

Quiz: suppose the class has 4 attractive people in it, and you want to meet them all for tea and cake. They don't know each other, so you plan 4 one-to-one meetups. These can be in any order (!) but each time, you scratch one name off your list of 4. What is the correct formula for this, (i) P(4,1)*P(4,1)*P(4,1)*P(4,1),
(ii) P(4,1)*P(3,1)*P(2,1)*P(1,1),
(iii) C(4,1)*C(4,1)*C(4,1)*C(4,1), or
(iv) C(4,1)*C(3,1)*C(2,1)*C(1,1)?

Reminder of a Definition:

C(n,k) = $\binom{n}{k} = \frac{n!}{(k!)(n-k)!} = \frac{\mathrm{P}(n,k)}{k!}$ Eq. 5

$\binom{4}{1} = \frac{4!}{(1!)(3!)} = 4$

But notice:

$\mathrm{P}(\mathrm{n},\mathrm{k}) = \frac{n!}{(n-k)!}$ So when k=1, the formula for C( ) reduces to the P( ) formula.

$\mathrm{P}(4,1) = \frac{4!}{(4-1)!} = 4!/3! = 4$

## << MORE CHALLENGING MARKET-RESEARCH QUESTIONS >>

<<NOTE: IF SOMETHING SEEMS IMPOSSIBLE, IT IS>>

Question 1 A category is a type of item, also called a line of stock. Looking at our external warehouse's 200 categories (or holdings), how many unique ways can we group any 40 of them into an aisle for customers or staff to collect them? _____

Question 2 A smaller store, Fernley's Furniture, has only 50 categories in stock. How many unique 40-category displays are there for Fernley's to choose from? _____

Question 3 Shop Aisle Task
The aisle of a shop or a storage area like Ikea can hold up to 40 large categories (20 per side) in order A1, A2, ... A20, B1, B2, B3,... B20. How many ordered options are there for holding an itemset of 40 items? _____

# MULTINOMIAL COEFFICIENTS

## (not used in Association Rules)

Suppose you plan to meet at a teashop were the tables hold 4 people. So you can meet up to 3 of the students each time, along with yourself. You decide you can widen your net and meet 15 students this semester in the teashop.

Each time, you plan to have 3 students come along. And you tick them off your list afterward. So, what is the total number of possible combinations (hint) of students that would meet your requirement that all 15 get met?

ANSWER:

HINT: See Libretexts, Multinomial Coefficients. With two choices made, the formula is:

$$Multinomial\binom{n}{k}\binom{n-k_1}{k_2} = \frac{n!}{k_1(n-k_1)!}\frac{n!}{k_2(n-(k_1+k_2))!} = \frac{n!}{k_1!(k_2!)(k_3!)} \qquad \text{Eq. 6}$$

In general, this gets converted into a new notation:

$$Multinomial\binom{n}{k_1,k_2,\ldots,k_r} = \frac{n!}{k_1!(k_2!)\ldots(k_r!)} \qquad \text{Eq. 7}$$

The problem is optional, because in association-rules we usually consider combinatorial choice from binary variables. So, C( ) is used, but it is not multinomial choice. In the future, if you are in a job interview, you may need to mention that you know the P( ) and C( ) as well as the multinomial choice algebra. Otherwise, you may appear confused. Yet it's really tricky.

Query: Once we assign all the invitees to the Tea & Cake dates, is the full list an itemset?

By Wendy Olsen 2024

Quick Quiz.

Q1 Which has/have distinct items:

{itemset, a permutation, a combination}

Q2 What mathematical rule is used for license plates to mix alpha and numeric chunks?

Q3 What is a factorial?

Q4 Let's choose 2 student reps from 30 students. How many distinct pairs are possible?

Q5 A buyer chooses 5 product lines out of 7 available. How many options (itemsets) are there to decide from?

Q6 What is an itemset?

Q7 Look at example 1.2 in 2.3 Combinations from Libretext. Is the answer $2^n$? What is it? These are baskets.

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Answers

Anagram task ‘Angleford’

Formula: P(9,1)*P(8,1)*P(7,1)*P(6,1)*P(5,1)*P(4,1)*P(3,1)*P(2,1)*P(1,1)

Insert your numbers: $\frac{9!}{8!}\frac{8!}{7!}\frac{7!}{6!}\frac{6!}{5!}\frac{5!}{4!}\frac{4!}{3!}\frac{3!}{2!}\frac{2!}{1!}\frac{1!}{1} = 9!$

=362,880

Calculation:

You would use a spreadsheet or calculator. You can also use any internet window. Type it like this: =9!

If that doesn’t work, you do it like this =9*8*7*6*5*4*3*2

Using R, check it. Your tutor will help you.

Second Anagram Task:

Note: to format the total number as "1,133,836,704,000", see R code.

The best anagram is simply gee.

License Plate Problem

The license plate problem is a repeat application of the combinations algebra. Repeating means multiplying. We do it group by group, group of 3, group of 2, and 2 groups of 1.

=24,219,000

Answer: The list of 24,219 thousand license plates is an itemset - Yes.
Answer: There are 9 distinct letters. The list of permutations far exceeds the number of combinations of the letters in Angleford.

Quiz: (iv) is correct.

Market Research Questions:

Question 1 Our category list does not count the stock by holdings in integers, eg we have 40 of category 1, 16 left of category 2. We ignore the counts. Instead, P(200, 40) is 200! / 60!. In R the result is so large, it is infinite.

Question 2 A smaller store, Fernley’s Furniture, has only 50 categories in stock. How many unique 40-category displays are there for Fernley’s to choose from? 10,272,278,170

Question 3 Shop Aisle Task

The aisle of a shop or a storage area like Ikea can hold up to 40 large categories (20 per side) in order A1, A2, … A20, B1, B2, B3,… B20. How many ordered options are there for holding this itemset? This last query exceeds R's capacity as the number is so large.

Tea & Cake query: Once we assign all the invitees to the Tea & Cake dates, is the full list an itemset? Yes.

Answers:

Q1 all three are itemsets

Q2 the product rule

Q3 a product of k, k-1, k-2, … 1.

Q4 335

Q5 21

Q6 a group of distinct, unique entities.